

Criteria for selecting EMC test instruments

1. The kind of task the user wants to perform. Requirements vary depending on the industry or application.
2. Working environment. If the equipment is for use in field, it needs to be heavy-duty, waterproof, rugged and portable. A benchtop instrument can perform delicate, precise measurements indoors.
3. Comprehensive coverage of test standard requirements so that the user can perform test to comply several standards using a single instrument.
4. Programmability and user friendliness of the test equipment.
5. Upgradability of the test equipment, in accordance with the changing standards or latest releases of the instrument, and local technical support available.

predict whether the given object will pass the emission test or not. For troubleshooting, high accuracy is not necessary, so you may go for low-cost probes or make your own current probes.

Antenna. A simple TV antenna positioned one or two metres away from the DUT would pick up harmonic signals. Connect it to the spectrum analyser to assess radiated emission. Changes below 2dB could be considered as measurement or setup error. Fixing the antenna in place using tape will reduce additional variables in measurement.

Preamplifier. A broadband preamplifier is used when the signal measured by the probes (particularly the smaller H-field probes) needs to be boosted in order to observe changes in emission levels.

ESD simulator. Electrostatic discharge (ESD) simulator or ESD gun generates ESD pulses that help verify the immunity of a particular device to static electricity discharges. While an actual ESD simulator can be expensive, you can create a rather simple ESD generator using a zipper storage bag with several coins inside. Shake this bag near the circuit to get several volts of ESD pulses at rise times of about 100 picoseconds.

ESD detector. Issues like loss of data and unusual circuit reset can be due to ESD in the circuit. Commercial ESD detectors are available in the market for higher accuracy applications. But an AM broadcast radio tuned off-station can act as a good ESD detector. If it has an FM receiver, you can even tune in product harmonics from radiated emissions.

Radiated immunity testing. Nothing can replace the test lab equipment for accurate radiated immunity levels. But a variable-RF signal generator

would allow the simulation of radiated immunity test to some degree and give you an idea of whether the object is immune or not. Once the susceptible region is identified, you can implement potential fixes. This can save time and money as opposed to performing troubleshooting at test labs.

Other contents. Some of the other things include digital multimeter, screwdriver kit, pencil soldering iron, solder, SMA (subminiature version A) connector wrench, flash light, magnifier, tweezers, wires of different lengths and sizes, 10dB and 20dB attenuators, aluminium foils, coaxial adaptors, insulation tape, copper tape, measuring tape, EMI gaskets, ferrite chokes, common-mode chokes, and a range of I/O cables, BNC coaxial cable, SMA coaxial cable, resistors, capacitors and inductors.

"At a compliance lab level, the list of test instruments and test infrastructure can be exhaustive," says Rajneesh Raveendran, the project manager of Tarang Test Lab, Wipro. But to get a product certified in accordance with the appropriate and latest test standards you always need to seek help of an EMC testing lab. "Semi/fully anechoic chamber, EMI receivers, antennae and antenna masts, turntable, signal generators, RF amplifiers, line impedance stabilisation networks, coupling/decoupling networks, RF current probes, various immunity test generators, etc are provided to cater to the various radiated and conducted emissions/immunity," adds Rajneesh.

Three design elements that influence testing

The size of the DUT decides the measurement method and test duration for

radiated emission and radiated immunity testing. As size of the object increases, the mechanical structure needs a good number of additional screws and holes to fix it. The additional slots and angles that come up to support the structure act as source of radiation, increasing the challenge faced by the engineer during compliance testing.

The complexity of its antenna pattern also increases with the size of the test object. "The antenna pattern complexity affects the number of orientations necessary to determine minimum immunity and maximum emission," says Vishal Gupta, application expert at Agilent Technologies India. This also has implications for EMC standards.

If your system's power consumption increases, it might result in the system requiring an active cooling system. This, in turn, demands for ventilation, creating openings that cause the radiation levels to increase, making it difficult to identify the source of radiation.

Similarly, with more number of cables, the tests will have to be done on each cable. Hence, for larger devices, more exposure is needed to measure gauge susceptibility and emissions. Moreover, the turntable or test chamber needs to be large to accommodate bigger equipment.

Very-near-field technique of EMC testing

The terms far-field, near-field and very-near-field describe the fields around an antenna. Near-field is less than one wavelength (λ) from the antenna and far field begins at a distance of 2λ and beyond. With the right implementation, any antenna can be successfully measured on either a near-field or far-field range. Ideally, far-field ranges are a better choice for lower frequency antennae and where simple pattern-cut measurements are required, and near-field ranges are better for higher frequency antennae and where complete pattern and polarisation measurements are required. The near-field is generally divided into two areas, the reactive and the radiative. The reactive region is what we call the very-near-field.